# EIGSEP Memo: Constraints on Ground Reflections for Global 21cm Measurements via $S_{11}$ Autocorrelation

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March 27, 2026

## 1 Introduction

As part of the EIGSEP experiment's field testing in Marjum Canyon, we performed antenna reflection ($S_{11}$) measurements on July 19, 2025 to characterize the local electromagnetic environment. The antenna was suspended at a height of 100 m to isolate terrestrial reflections at large delays. This memo analyzes the autocorrelation of the $S_{11}$ voltage spectrum to distinguish between primary interference (direct signal with reflection) and secondary interference (reflection with reflection).

## 2 Measurement and Analysis

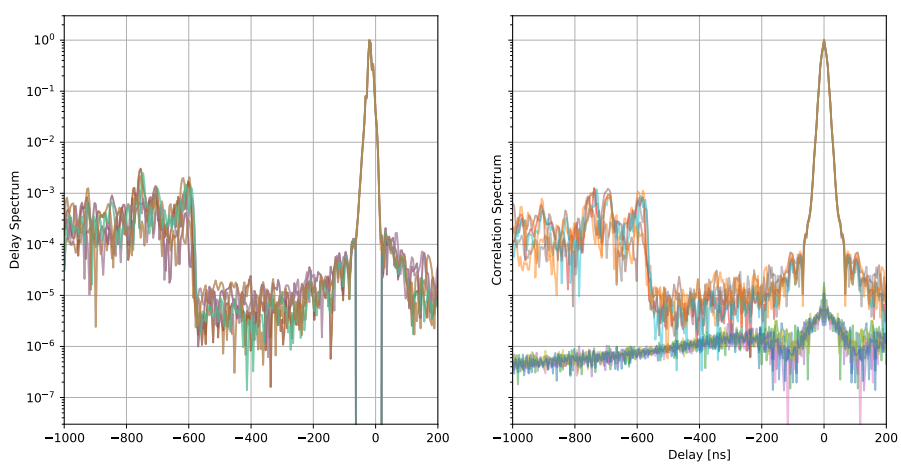


Figure 1: **S11 Delay and Autocorrelation Analysis for Marjum Canyon. Left (Delay Spectrum):** The normalized volage delay spectrum of the antenna $S_{11}$ at 100 m suspension. The primary peak at 0 ns is the intrinsic antenna reflection, while peaks at $\approx \pm 600$–700 ns correspond to ground reflections. **Right (Autocorrelation Spectrum):** Comparison of $|V(\nu)|^2$ for the *Full* $S_{11}$ (upper) and *Filtered* (lower) cases. The filtered case removes the 0-delay peak before squaring to isolate reflection-to-reflection correlations ($\epsilon \star \epsilon$). The difference in power levels suggests that the spectral ripples are dominated by the primary interference of reflections with the direct signal ($A_0 \star \epsilon$), rather than mutual interference between reflections.

Measurements were collected using a Vector Network Analyzer (VNA) from 1–500 MHz. The voltage reflection coefficient $V(\nu)$ can be modeled as:

$$V(\nu) = A_0 + \sum_i \Gamma_i e^{-j2\pi\nu\tau_i} \tag{1}$$

where $A_0$ represents the intrinsic antenna mismatch (at 0-delay) and $\Gamma_i$ represents ground reflections at delays $\tau_i$. The power spectrum measured by the VNA is proportional to the square of the voltage, $|V(\nu)|^2$:

$$|V(\nu)|^2 = |A_0|^2 + \underbrace{2\mathrm{Re}\left(A_0 \sum_i \Gamma_i^* e^{j2\pi\nu\tau_i}\right)}_{\text{Primary (Antenna} \star \text{Reflection)}} + \underbrace{\left|\sum_i \Gamma_i e^{-j2\pi\nu\tau_i}\right|^2}_{\text{Secondary (Reflection} \star \text{Reflection)}} \tag{2}$$

Figure 1 compares the full autocorrelation spectrum against a version where the 0-delay reference ($A_0$) is removed via a windowing mask ($\pm$40 ns) before squaring. The resulting comparison shows that the full $S_{11}$ spectrum is everywhere at least one order of magnitude larger than the filtered spectrum containing only reflections outside $\approx$ 40 ns.

## 3 Imperfect Reciprocity

While these VNA measurements are active (transmitting from the antenna), the results directly constrain the passive case of sky-sourced signals. By reciprocity, the reflection coefficient $\Gamma$ that characterizes the return of an antenna-sourced signal is the same coefficient that determines the magnitude of a reflected sky signal entering the antenna.

In a passive observation, the total received power $P(\nu)$ from a sky signal $S_{\mathrm{sky}}$ is:

$$P(\nu) \propto |S_{\mathrm{sky}}|^2 \left(1 + 2\mathrm{Re}(\Gamma e^{-j2\pi\nu\tau}) + |\Gamma|^2\right) \tag{3}$$

Our $S_{11}$ data confirms that $|\Gamma|^2 \ll 2\Gamma$. This demonstrates that the environment is sufficiently non-reflective that reflections correlating with reflections (multipath) are negligible.

However, the actual cosmology signal path would be from the sky to the antenna, and from the sky to the terrain and then reflecting to the antenna. The true reciprocal signal path excited by the VNA must end on the sky, so it should be mentioned that the geometries of these reflections do not in detail match those that will influence our science measurements.

## 4 Significance for 21cm Cosmology

The primary systematic challenge in global 21cm measurements is the high-frequency spectral ripple caused by ground reflections. This result indicates that:

- **Simplified Modeling:** The canyon floor does not act as a high-Q cavity; we only need to model first-order spectral interference between the direct and reflected sky signal.
- **Spectral Purity:** Because secondary interference is suppressed, the resulting ripple is a clean sinusoid at a delay $\tau$, which can be more easily decoupled from the broad, global 21cm absorption feature.

# 5 Conclusion

The suspension of the EIGSEP antenna at 100 m in Marjum Canyon reveals an environment dominated by primary signal-to-reflection interference. This provides a robust constraint on terrestrial systematics, allowing the analysis to focus on the interference of the reflected sky signal with the direct signal as the primary source of spectral contamination.